Akoizo Original Practice - Form 1 - Answer key

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which is most operational?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Time in seconds between two fixed marks

B. How good the motion is

C. Whether the object seems fast

D. Interesting behavior

Correct: A

2. (2 points) A fixed +0.5 s offset is…

A. A new independent variable always

B. No error

C. Systematic bias

D. Only random scatter

Correct: C

3. (2 points) Which is a direct qualitative observation?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Friction definitely increased

B. The hypothesis is true

C. The cause was gravity alone

D. The object wobbled

Correct: D

4. (2 points) A missing trial should be fabricated to complete the table?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only if software predicts it

B. No

C. Yes

D. Only if the hypothesis is clear

Correct: B

5. (2 points) A source of error explanation should include…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Only a list of tools

B. Only an apology

C. How it changes the response

D. Only the word “human”

Correct: C

6. (4 points) Write a directional ramp hypothesis.

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: Increasing release height will reduce defined transit time under comparable conditions because available potential energy increases. (4 points).

7. (6 points) Solve this case and explain your reasoning: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.

reasoning: Calculate (2 + 3 + 4)/3 = 3 g.
Calculate (5 + 6 + 7)/3 = 6 g.
Compare the means while acknowledging the 2 g range in each group. (3 points).

conclusion: Doubling area increased mean absorbed mass from 3 g to 6 g (3 points).

8. (4 points) Distinguish SD and standard error.

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: SD describes individual-value spread; standard error estimates mean precision under valid independence assumptions. (4 points).

9. (4 points) Why use several levels?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: They reveal the response pattern over a range rather than only a single pairwise difference. (4 points).

10. (2 points) Define “Mean” in the context of Experimental Design.

answer: Sum divided by number of measurements. (2 points).

11. (4 points) Why can precise measurements be inaccurate?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: A systematic offset can shift all tightly clustered readings away from the reference. (4 points).

12. (6 points) Solve this case and explain your reasoning: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

reasoning: Choose release height as the independent variable with defined levels.
Define response as time between two fixed marks, with units and timing endpoints.
Hold object, ramp surface, release method, and travel distance consistent and replicate each level. (3 points).

conclusion: A narrow measurable comparison supports a reproducible experiment. (3 points).

13. (6 points) Solve this case and explain your reasoning: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

reasoning: The offset is systematic.
Averaging more readings can reduce some random variation but retains the 0.5 s offset.
Calibrate or correct using a justified reference method and document the change. (3 points).

conclusion: Replication alone cannot remove a known systematic offset. (3 points).

14. (2 points) Define “Qualitative data” in the context of Experimental Design.

answer: Descriptive observations rather than the primary numerical measurement. (2 points).

15. (2 points) Define “Dependent variable” in the context of Experimental Design.

answer: Measured response. (2 points).

16. (2 points) Define “Systematic bias” in the context of Experimental Design.

answer: Consistent or structured distortion. (2 points).

17. (2 points) Define “Residual” in the context of Experimental Design.

answer: Observed value minus model prediction. (2 points).

18. (2 points) Define “Confounding” in the context of Experimental Design.

answer: A competing factor systematically linked with the intended factor. (2 points).